

Discrete-Time Net Capture Against an Intruder Using Multiple UAVs

JIANAN WANG ^{ID}, Senior Member, IEEE
 LONGZE ZHAO ^{ID}
 KEWEI XIA ^{ID}
 CHUNYAN WANG ^{ID}
 Beijing Institute of Technology, Beijing, China

This article investigates the problem of aerial capture using a net manipulated by multiple unmanned aerial vehicles (UAVs). To guarantee the interception of the intruder in practical sampling instants, a 3-D discrete-time proportional navigation-like guidance law is proposed for the geometric center of the net. Meanwhile, to capture the intruder within a considerably large interception area, a net-plane orienting strategy is formulated for each UAV. This strategy guarantees the determination of the desired positions for each UAV and designs a UAV controller considering disturbance compensation to drive the UAVs to track the desired positions, which ensures that the net plane becomes orthogonal to the guidance motion precisely at the instant of net intercepts the intruder. Finally, numerical simulations are conducted to illustrate the efficacy of the proposed approach.

Received 8 September 2023; revised 6 March 2024 and 10 September 2024; accepted 15 September 2024. Date of publication 19 September 2024; date of current version 14 April 2025.

DOI. No. 10.1109/TAES.2024.3464570

Refereeing of this contribution was handled by A. Ratnoo.

This work was supported by the National Natural Science Foundation of China under Grant 62273043 and Grant 61933002.

Authors' address: Jianan Wang, Longze Zhao and Chunyan Wang are with the School of Aerospace Engineering, Beijing Institute of Technology, Beijing, 100081, China, E-mail: (wangjianan@ieee.org; zhaolongze@bit.edu.cn; chunyan.wang@bit.edu.cn); Kewei Xia is with the Advanced Research Institute of Multidisciplinary Sciences, Beijing Institute of Technology, Beijing, 100081, China, E-mail: (kwxia@bit.edu.cn). (Corresponding author: Kewei Xia.)

0018-9251 © 2024 IEEE

NOMENCLATURE

$(\cdot)(k)$	Value of $(\cdot)$ at the k th sampling instant.
$(\cdot)^k$	Location of $(\cdot)$ at the k th sampling instant.
$\mathbf{a}_C$	Acceleration vector of C .
$\mathbf{a}_{i,d}$	Desired acceleration vector of P_i .
$\mathbf{r}_C$	Position vector of the geometric center of the net.
$\mathbf{r}_i$	Position vector of the i th net-carrying UAV.
$\mathbf{r}_T$	Position vector of the aerial intruder.
$\mathbf{r}_{i,d}$	Desired position of the i th net-carrying UAV.
$\mathbf{r}_{VL}$	Virtual line-of-sight (VLOS) vector.
$\mathbf{v}_C$	Velocity vector of C .
$\mathbf{v}_{i,d}$	Desired velocity vector of P_i .
$\mathbf{v}_i$	Velocity vector of P_i .
χ_i	Heading angle of the i th net-carrying UAV.
γ_i	Flight-path angle of the i th net-carrying UAV.
ϕ_{bi}	Bank angle the i th net-carrying UAV.
θ_{VL}, ψ_{VL}	Azimuth angle and elevation angle of VLOS.
C	Location of the geometric center of the net.
D_{gi}	Drag of the i th net-carrying UAV.
g	Gravitational coefficient.
K	Guidance gain.
L	Original length of the edge of the captrue net.
L_{fi}	Lift of the i th net-carrying UAV.
m_i	Mass of the i th net-carrying UAV.
P_i	Location of the i th net-carrying UAV.
T	Location of the aerial intruder.
t_k	k th sampling instant.
T_{hi}	Thrust of the i th net-carrying UAV.
V_i	Ground speed of the i th net-carrying UAV.

I. INTRODUCTION

With the advancement of unmanned aerial vehicle (UAV) technology, there is a growing emphasis on security and defense against emerging threats. When unauthorized drones intrude into airspace, it becomes imperative to swiftly capture or neutralize them. A potential approach to this problem is to use a capture net tensioned by multiple UAVs. Compared with destroying the intruder with missiles, this approach offers a cost-effective and practical solution, and is suitable for a wide range of environments including densely populated urban areas and open rural terrain. Also, the UAV team can be reused even after it accomplishes the mission [1], [2], [3].

Several aspects of the aerial net capture problem were discussed in existing literature [4], [5], [6], [7], [8], [9], [10]. Diehl et al. [4] addressed the concept of interceptor UAV, which is a quadrotor carrying a rigid capture net, but there is a lack of analysis and simulation to support this concept. Motivated by the idea of conceptualizing the geometric center of the capture net as the virtual leader, an approach to catching a drone by two UAVs was developed in [5] and [6], but the attitude of the net cannot be adjusted during the capture process in this approach. In [7], a possible solution with four UAVs carrying a dragnet was considered. In this scenario, the aerial intruder moves toward a certain

goal point so that the dragnet-carrying UAVs can capture it by guidance algorithms under the assumption that the intruder's velocity can be measured in real time. Kim et al. [8] introduced a discrete 3-D controller for enabling a team of net-carrying UAVs to capture an intruder with variable velocity. This controller addresses the capture problem by maintaining the line-of-sight (LOS) pair stationary. However, this approach cannot control the rotate angle of the net. Chakravarthy et al. [11] investigated a collision cone-based guidance law for n UAVs to pursue a swarm of target on a plane. In [10] and [12], the collision cone-based strategy was extended to 3-D space for multiple net-carrying UAVs to intercept the intruder swarm with an appropriate orientation of the net to complete the capture mission with satisfactory performance. Although the aforementioned approach have been proven kinematic feasible, the dynamics of UAV is typically nonnegligible.

To accomplish the interception objective, applying guidance law to the net center is a reasonable approach [13]. Many scholars had developed various guidance laws for missiles and UAVs to intercept stationary, nonmaneuvering or maneuvering aerial intruders in continuous process [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. In [14], the capture region of the proportional navigation (PN) guidance law was analyzed by the geometric approach. The validity of the pure PN (PPN) for maneuvering target was proved in [15] and [16]. The guidance laws for UAVs and missiles with acceleration saturation constraint were proposed in [17], [18], and [19]. In [20], a terminal angle constrained cooperative guidance law for multiple missiles against maneuvering target was derived. Dong et al. converted the 3-D guidance problem into a 2-D one by introducing the concept of the shaping plane and the pursuing plane [23]. Jana et al. [24] presented a methodology to design a 3-D cooperative guidance law for a dual interceptor system carrying an elongated warhead. However, in practical applications, the signal transmission may be delayed because of the distance between the interceptor and the target [25]. Besides, the missiles and UAVs can only change their guidance commands at sampling instants. Therefore, some researchers approached the guidance problem as a discrete process-related challenge [26], [27], [28], [29]. In [27], a kind of discrete guidance law was proposed based on sliding mode control method, but the approach is hard to be generalized to 3D space. Gao converted the PN into discrete form by difference equations [30]. Nevertheless, it is not real-time and cannot handle the case where the moving directions of the interceptor and the target are not in the same plane.

During flight of UAV, the disturbance generated by the net tension is typically significant and cannot be ignored. Therefore, it is necessary to compensate it in the UAV controller. A robust PID controller was proposed for autopilot in [31], primarily compensating for disturbance arising from parameter uncertainties. Lee et al. designed a controller for a constantly spinning imbalanced quadrotor based on disturbance observers, achieving good control performance in experiments [32]. A dynamics estimator based on a

first-order filter that can effectively estimate disturbance in scenarios, such as landing under strong wind for quadrotors was presented in [33], [34]. Since only one filter parameter is involved in the estimator, parameter tuning can be significantly simplified. However, this approach cannot be directly applied to discrete systems.

Note that the desired positions of the net-carrying UAVs need to be solved in real time to obtain an accurate position of the net center. Furthermore, the orientation of the net is determined by the positions of the UAVs as well. Motivated by aforementioned discussions, we investigate the net capture problem by employing a 3D discrete guidance law and a net-plane orientating strategy. Their joint efforts should be made to solve the problem of interception on maneuvering aerial intruder by multiple net-carrying UAVs. The 3D discrete guidance law proposed in this study employs the geometric center of the net as a "virtual interceptor", accomplishing target interception by solely relying on the position error vectors between each net-carrying UAV and the target using the proposed PN-like guidance law. This approach effectively overcomes the challenge posed by the inability to install sensors on the capture net. It directly tackles the challenges associated with the absence of LOS angles, LOS angular velocities, distance measurements, and their derivatives, which often lead to the ineffectiveness of traditional guidance laws in such scenarios. The net orientating strategy is outlined as follows: By utilizing the desired position of the net center provided by the guidance law, the desired positions of the net-carrying UAVs can be determined using proposed geometric approach. Subsequently, a discrete UAV controller is designed, where the net tension disturbance is compensated by a first-order filter based dynamics estimator. This ensures that the plane formed by the net-carrying UAVs, commonly named as the "net plane", ultimately becomes orthogonal to the interceptor's motion direction.

The rest of article is organized as follows. Section II introduces the problem statement of this article. Section III presents the discrete 3D guidance law to ensure the net center can intercept intruder successfully. Section IV presents an approach to adjusting orientation of the net by controlling each net-carrying UAV to accomplish capture task. Section V presents some simulation results to verify the performance of the proposed method. Finally, Section VI concludes this article.

II. PROBLEM STATEMENT

A. Problem Formulation

A scenario involved with a capture net carried by n UAVs and an aerial intruder is considered in this work (see Fig. 1). The desired configuration of net-carrying UAVs is a regular polygon with n sides, thereby ensuring the net remains taut and retains its planar structure. Furthermore, throughout the interception process, the line connecting the two lowest altitude UAVs remains parallel to the horizontal plane, as illustrated in Fig. 2. The guidance objective is to maneuver the net by UAVs so as to reduce the distance between C and

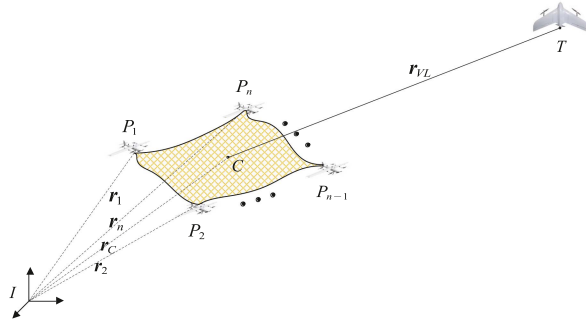


Fig. 1. Schematic of n UAVs transporting a net to intercept an aerial intruder.

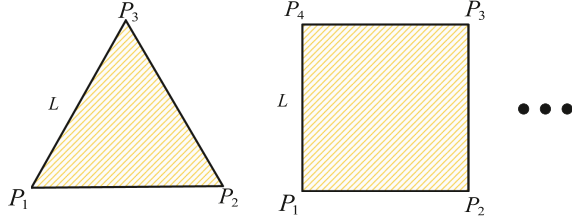


Fig. 2. Schematic of geometrical configuration of n UAVs.

T with the additional requirement that rotate the net plane to be perpendicular to C 's maneuvering direction.

ASSUMPTION 1 The net-carrying UAVs are capable of measuring their own global positions by the onboard sensors, and the ground radar can measure the position of the intruder.

B. UAV Model

This article utilizes a three degree-of-freedom aircraft model [35] to characterize the motion of the net-carrying UAVs. The model assumes that the thrust aligns with the velocity vector. Furthermore, it is assumed that the Earth is flat and nonrotating. With these premises, the equations of the i th net-carrying UAV can be described by

$$\dot{x}_i = V_i \cos \gamma_i \cos \chi_i \quad (1a)$$

$$\dot{y}_i = V_i \cos \gamma_i \sin \chi_i \quad (1b)$$

$$\dot{z}_i = V_i \sin \gamma_i \quad (1c)$$

$$\dot{V}_i = \frac{T_{h_i} - D_{g_i}}{m_i} - g \sin \gamma_i \quad (1d)$$

$$\dot{\gamma}_i = \frac{g}{V_i} (n_{g_i} \cos \phi_{b_i} - \cos \gamma_i) \quad (1e)$$

$$\dot{\chi}_i = \frac{L_{f_i} \sin \phi_{b_i}}{m_i V_i \cos \gamma_i} \quad (1f)$$

where the control variables in the UAV model are the load factor $n_{g_i} = L_{f_i}/(m_i g)$, controlled by the elevator; the bank angle ϕ_{b_i} , controlled by a combination of rudder and ailerons, and the thrust T_{h_i} is controlled by the throttle.

Differentiating (1a)–(1c), and substituting for V_i , γ_i , and χ_i from (1d)–(1f), and introducing the external disturbance, the model can be simplified into the form of a second-order

integrator as

$$\ddot{x}_i = U_{x_i} + d_{x_i} \quad \ddot{y}_i = U_{y_i} + d_{y_i} \quad \ddot{z}_i = U_{z_i} + d_{z_i} \quad (2)$$

where d_{x_i} , d_{y_i} , and d_{z_i} represent the components of external disturbance $\mathbf{D}_i$ in the three coordinate directions, respectively. U_{x_i} , U_{y_i} , and U_{z_i} are the components of new pseudocontrol variables $\mathbf{U}_i$. The relationships between these pseudocontrol variables and the actual control variables are expressed as

$$\begin{cases} \phi_{b_i} = \tan^{-1} \left[\frac{U_{y_i} \cos \chi_i - U_{x_i} \sin \chi_i}{\cos \gamma_i (U_{z_i} + g) - \sin \gamma_i (U_{x_i} \cos \chi_i + U_{y_i} \sin \chi_i)} \right] \\ n_{g_i} = \frac{\cos \gamma_i (U_{z_i} + g) - \sin \gamma_i (U_{x_i} \cos \chi_i + U_{y_i} \sin \chi_i)}{g \cos \phi_{b_i}} \\ T_{h_i} = \left[\sin \gamma_i (U_{z_i} + g) + \cos \gamma_i (U_{x_i} \cos \chi_i + U_{y_i} \sin \chi_i) \right] \\ m_i + D_{g_i}. \end{cases} \quad (3)$$

C. Capture Net Model

The capture net is assumed to be stretchable. The mass of the net is neglected. The external disturbance caused by the net (which is not available for net-carrying UAVs) is modeled as a semispring damper model [36]

$$\mathbf{D}_i = \begin{cases} \sum_{j=1, j \neq i}^n k_c (r_{ij} - L) \frac{\mathbf{r}_{ij}}{r_{ij}} + c \dot{\mathbf{r}}_{ij}, & r_{ij} > L \\ 0, & r_{ij} \leq L \end{cases} \quad (4)$$

where $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, $r_{ij} = \|\mathbf{r}_{ij}\|$, k_c and c represent the elastic coefficient and damping coefficient of the capture net, respectively. The end strings of the capture net are attached to the center of mass of the UAVs. Furthermore, the length of the string is long enough to ensure that the change of UAVs' attitude will not interfere with the net.

ASSUMPTION 2 [37] The change rate of external disturbance $\mathbf{D}_i$ is bounded, i.e., $\|\mathbf{D}_i(k+1) - \mathbf{D}_i(k)\| \leq \sigma_d$, with an unknown positive constant σ_d .

D. Geometric Model

The net center's position is determined as

$$\mathbf{r}_C = \frac{1}{n} \sum_{i=1}^n \mathbf{r}_i. \quad (5)$$

Therefore, the VLOS vector can be expressed as

$$\mathbf{r}_{VL} = \mathbf{r}_T - \mathbf{r}_C. \quad (6)$$

III. 3-D DISCRETE-TIME PN-LIKE GUIDANCE LAW

The engagement geometry that governs the discrete-time kinematics is illustrated in Fig. 3. The approach employed in this section to address the guidance problem is based on the idea of driving C to strike T . The vector denoted by $C^k T^{k-1}$ corresponds to $\mathbf{r}_{VL}(k)$, and its magnitude is denoted as $r_{VL}(k)$. The primary goal of this guidance problem is to reduce $r_{VL}(k)$ until C intercepts T . The azimuth angle $\theta_{VL}(k)$ and elevation angle $\psi_{VL}(k)$ of $\mathbf{r}_{VL}(k)$ can be

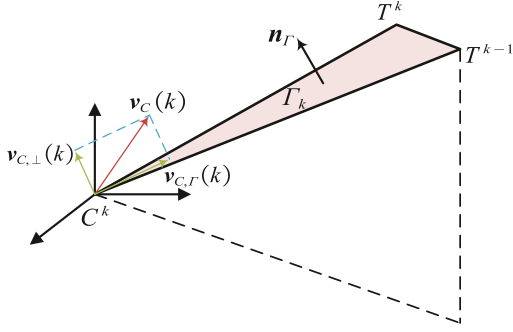


Fig. 3. Schematic of guidance geometry.

expressed as

$$\begin{cases} \theta_{VL}(k) = \arcsin\left(\frac{x_{VL}(k)}{\|r_{VL}(k)\|}\right) \\ \psi_{VL}(k) = \arccos\left(\frac{z_{VL}(k)}{\|r_{VL}(k)\| \cos \theta_{VL}(k)}\right). \end{cases} \quad (7)$$

DEFINITION 1 As depicted in Fig. 3, the plane established by the intersection of $C^k T^{k-1}$ and $C^k T^k$ is designated as the instantaneous guidance plane, which is denoted as Γ_k .

Resolve $v_C(k)$ into two constituent components: 1) $v_{C,\Gamma}(k)$ lying within Γ_k , and 2) $v_{C,\perp}(k)$ orthogonal to Γ_k . The unit normal vector of the plane Γ_k is expressed as

$$n_{\Gamma}(k) = \frac{r_{VL}(k) \times r_{T^{k-1}T^k}}{\|r_{VL}(k) \times r_{T^{k-1}T^k}\|} \quad (8)$$

where $r_{T^{k-1}T^k}$ represents the vector from T^{k-1} to T^k . Then, $v_{C,\Gamma}(k)$ and $v_{C,\perp}(k)$ satisfy

$$\begin{cases} v_{C,\perp}(k) = (v_C(k) \cdot n_{\Gamma}(k)) n_{\Gamma}(k) \\ v_{C,\Gamma}(k) = v_C(k) - v_{C,\perp}(k). \end{cases} \quad (9)$$

Then, the 3-D guidance problem can be separated into a 2-D guidance problem at Γ_k and a control problem which aims to reduce the magnitude of $v_{C,\perp}(k)$ to zero.

A. Two-Dimensional Guidance Strategy on Γ_k

The issue in this study is described as a discrete PN guidance problem at Γ_k based on geometric method (see Figs. 4 and 5). Define $\phi_{\Gamma}(k)$ as the angle between $r_{VL}(k)$ and $v_{C,\Gamma}(k)$, which is calculated as

$$\phi_{\Gamma}(k) = \arccos\left(\frac{v_{C,\Gamma}(k) \cdot r_{VL}(k)}{\|v_{C,\Gamma}(k)\| \cdot \|r_{VL}(k)\|}\right). \quad (10)$$

Define

$$n_{\text{jud}} = \frac{v_{C,\Gamma}(k) \times r_{VL}(k)}{\|v_{C,\Gamma}(k) \times r_{VL}(k)\|} - \frac{r_{T^{k-1}T^k} \times r_{VL}(k)}{\|r_{T^{k-1}T^k} \times r_{VL}(k)\|}$$

as a judge vector for assessing whether the directions of $v_{C,\Gamma}(k)$ and $r_{T^{k-1}T^k}$ align on the same side of $r_{VL}(k)$ or not.

ASSUMPTION 3 The initial velocity direction of C is oriented toward the target, implying that when $k = 1$, the projection of C 's velocity onto the VLOS yields a positive value. Formally, this condition can be expressed as $\frac{r_{VL}(1) \cdot v_{C,\Gamma}(1)}{\|r_{VL}(1)\| \cdot \|v_{C,\Gamma}(1)\|} > 0$.

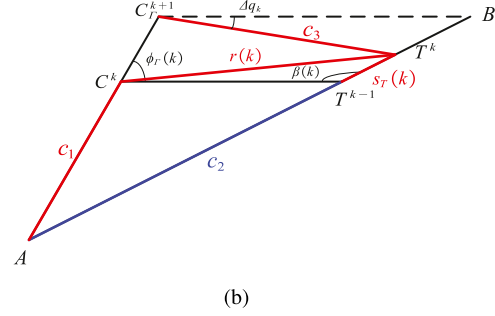
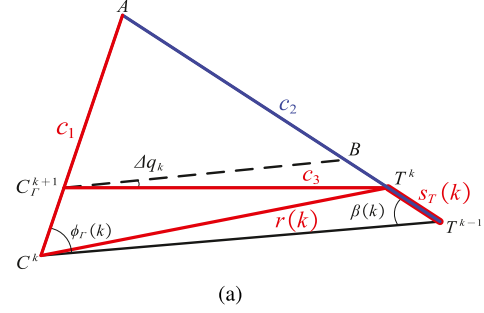


Fig. 4. Schematic of 2-D guidance problem under the case of $n_{\text{jud}} = 0$. (a) $\phi_{\Gamma}(k) < \pi - \beta(k)$. (b) $\phi_{\Gamma}(k) > \pi - \beta(k)$.

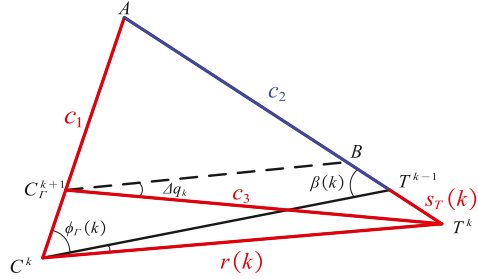


Fig. 5. Schematic of 2-D guidance problem under the case of $n_{\text{jud}} \neq 0$.

Based on the stipulations outlined in Assumption 3, the guidance problem can be separated into two scenarios based on the magnitude of n_{jud} , labeled as n_{jud} .

1) *Guidance Strategy Under $n_{\text{jud}} = 0$* : The directions of $v_{C,\Gamma}(k)$ and $r_{T^{k-1}T^k}$ coincide on the same side of $r_{VL}(k)$, as depicted in the corresponding geometric illustration in Fig. 4. Denote the projection of point C^{k+1} on Γ_k as point C_{Γ}^{k+1} . The length traveled by the net center on Γ_k during one sample interval is calculated as $s_C(k) = \|v_{C,\Gamma}(k)\| \cdot \Delta t$, where $\Delta t = t_k - t_{k-1}$. Apparently, the angle between $C^k C_{\Gamma}^{k+1}$ and $C^k T^{k-1}$ is $\phi_{\Gamma}(k)$. Extend the lines $C^k C_{\Gamma}^{k+1}$ and $T^{k-1} T^k$ to intersect at point A. Extend a parallel line from point C_{Γ}^{k+1} to the line $C^k T^k$ and let it intersect the line $T^k A$ at point B. Define $C^k T^k = r(k)$, $s_T(k) = \|r_{T^{k-1}T^k}\| = T^{k-1} T^k$, $AC^k = c_1$, $AT^{k-1} = c_2$, $C_{\Gamma}^{k+1} T^k = c_3$, $\angle T^k T^{k-1} C^k = \beta(k)$, and $\angle T^k C_{\Gamma}^{k+1} B = \Delta q_k$. Following the above definitions, one can get

$$\beta(k) = \arccos\left(\frac{r_{VL}^2(k) + s_T^2(k) - r^2(k)}{2r_{VL}(k)s_T(k)}\right). \quad (11)$$

If $\phi_\Gamma(k) < \pi - \beta(k)$, as shown in Fig. 4(a), one can obtain

$$\begin{cases} c_1 = \frac{r_{VL}(k) \cdot \sin \beta(k)}{\sin(\phi_\Gamma(k) + \beta(k))} \\ c_2 = \frac{r_{VL}(k) \cdot \sin \phi_\Gamma(k)}{\sin(\phi_\Gamma(k) + \beta(k))} \\ AC_\Gamma^{k+1} = c_1 - s_C(k) \\ AT^k = c_2 - s_T(k) \\ c_3 = \sqrt{\frac{AC_\Gamma^{k+1} + AT^{k2}}{+2AC_\Gamma^{k+1}AT^k \cos(\phi_\Gamma(k) + \beta(k))}} \end{cases} \quad (12)$$

Then, Δq_k can be formulated as

$$\Delta q_k = \phi_\Gamma(k) - \arccos \left(\frac{AC_\Gamma^{k+1} + c_3^2 - AT^{k2}}{2AC_\Gamma^{k+1}c_3} \right). \quad (13)$$

If $\phi_\Gamma(k) > \pi - \beta(k)$, as shown in Fig. 4(b), one can obtain

$$\begin{cases} c_1 = \frac{r_{VL}(k) \cdot \sin \beta(k)}{\sin(\phi_\Gamma(k) + \beta(k))} \\ c_2 = \frac{r_{VL}(k) \cdot \sin \phi_\Gamma(k)}{\sin(\phi_\Gamma(k) + \beta(k))} \\ AC_\Gamma^{k+1} = c_1 + s_C(k) \\ AT^k = c_2 + s_T(k) \\ c_3 = \sqrt{\frac{AC_\Gamma^{k+1} + AT^{k2}}{+2AC_\Gamma^{k+1}AT^k \cos(\phi_\Gamma(k) + \beta(k))}} \end{cases} \quad (14)$$

and thus, Δq_k is expressed as

$$\Delta q_k = \phi_\Gamma(k) + \arccos \left(\frac{AC_\Gamma^{k+1} + c_3^2 - AT^{k2}}{2AC_\Gamma^{k+1}c_3} \right) - \pi. \quad (15)$$

Then, one can convert the PN to discrete form. The guidance equation with respect to angle is equivalent as

$$\phi_\Gamma(k+1) = \phi_\Gamma(k) + K \cdot \Delta q_k. \quad (16)$$

As shown in Fig. 4, the coordinates of point A, denoted as $\mathbf{r}_A$, can be expressed as

$$\mathbf{r}_A(k) = \mathbf{r}_T(k-1) + \frac{c_2}{s_T(k)} \mathbf{r}_{T^{k-1}T^k}. \quad (17)$$

The coordinates of point C_Γ^{k+1} is denoted as $\mathbf{r}_{C,\Gamma}(k+1)$, and can be formulated as

$$\mathbf{r}_{C,\Gamma}(k+1) = \mathbf{r}_C(k) + \text{sgn}(\nu) \frac{s_C(k)}{c_1} (\mathbf{r}_A(k) - \mathbf{r}_C(k)) \quad (18)$$

where $\nu = \phi_\Gamma(k) + \beta(k) - \pi$.

2) *Guidance Strategy Under $n_{\text{jud}} \neq 0$* : The directions of $\mathbf{v}_{C,\Gamma}(k)$ and $\mathbf{r}_{T^{k-1}T^k}$ are at different side of $\mathbf{r}_{VL}(k)$. The geometrical relationship is depicted as Fig. 5, where $\beta(k)$ is calculated by

$$\beta(k) = \pi - \arccos \left(\frac{r_{VL}^2(k) + s_T^2(k) - r^2(k)}{2r(k)s_T(k)} \right). \quad (19)$$

Then, according to the geometric relationship, one can obtain

$$\begin{cases} c_1 = \frac{r_{VL}(k) \cdot \sin \beta(k)}{\sin(\phi_\Gamma(k) + \beta(k))} \\ c_2 = \frac{r_{VL}(k) \cdot \sin \phi_\Gamma(k)}{\sin(\phi_\Gamma(k) + \beta(k))} \\ \angle T^{k-1}C^kT^k = \arcsin \left(\frac{s_T(k) \cdot \sin \beta(k)}{r(k)} \right) \\ \angle C^{k+1}C^kT^k = \phi_\Gamma(k) + \angle T^{k-1}C^kT^k \\ c_3 = \sqrt{r(k)^2 + s_C^2(k) + 2r(k) \cdot s_C(k) \cos \angle C^{k+1}C^kT^k} \end{cases} \quad (20)$$

One can get an accurate Δq_k as

$$\Delta q_k = \arcsin \left(\frac{s_C(k) \sin \angle C_\Gamma^{k+1}C^kT^k}{c_3} \right). \quad (21)$$

The guidance equation is analogous in form to (16). Then, $\mathbf{r}_A$ and $\mathbf{r}_{C,\Gamma}(k+1)$ share the same form as (17) and (18) with $\nu > 0$.

REMARK 1 Under the constraint of *Assumption 3*, the angle guidance command (16) will drive $\phi_\Gamma(k)$ to converge to zero. That is, the direction of $\mathbf{v}_C(k)$ consistently strives to align itself with the VLOS vector.

The guidance command within Γ_k is designated as $\mathbf{a}_\Gamma(k)$. Denote the unit vector of $\mathbf{a}_\Gamma(k)$ as $\hat{\mathbf{a}}_\Gamma(k)$. It is apparent that $\hat{\mathbf{a}}_\Gamma(k)$ can be obtained by rotating $\hat{\mathbf{v}}_{C,\Gamma}(k)$ 90° counterclockwise around $\mathbf{n}_\Gamma$, and can be solved by

$$\begin{cases} \mathbf{R}_a(k) = \cos \frac{\pi}{2} \mathbf{I}_{3 \times 3} + \left(1 - \cos \frac{\pi}{2}\right) \mathbf{n}_\Gamma(k) \cdot \mathbf{n}_\Gamma^T(k) \\ \quad + \sin \frac{\pi}{2} \mathbf{n}_\Gamma^\times(k) \\ \hat{\mathbf{a}}_\Gamma(k) = \mathbf{R}_a(k) \hat{\mathbf{v}}_{C,\Gamma}(k) \end{cases} \quad (22)$$

where $\mathbf{n}_\Gamma^\times(k) = \begin{bmatrix} 0 & -n_z & n_y \\ n_z & 0 & -n_x \\ -n_y & n_x & 0 \end{bmatrix}$, with $\mathbf{n}_\Gamma(k) = [n_x, n_y, n_z]^T$. Then, according to (16), $\mathbf{a}_\Gamma(k)$ is calculated by

$$\mathbf{a}_\Gamma(k) = \frac{K \cdot \|\mathbf{v}_C(k)\| \Delta q_k}{\Delta t} \hat{\mathbf{a}}_\Gamma(k). \quad (23)$$

B. Control of $\mathbf{v}_C$

Define an error variable e_r to describe the deviation between $\mathbf{v}_C$ and Γ_k

$$e_r = (\mathbf{v}_C \cdot \mathbf{n}_\Gamma)^2. \quad (24)$$

By differentiating (24), one can obtain

$$\dot{e}_r = 2\mathbf{v}_C \cdot \mathbf{n}_\Gamma (\dot{\mathbf{v}}_C \cdot \mathbf{n}_\Gamma + \mathbf{v}_C \cdot \dot{\mathbf{n}}_\Gamma). \quad (25)$$

Then, the discrete-time form of (25) is expressed as

$$\frac{e_r(k) - e_r(k-1)}{\Delta t} = 2\mathbf{v}_C(k) \cdot \mathbf{n}_\Gamma(k-1) \left[\frac{\mathbf{v}_{C,d}(k+1) - \mathbf{v}_C(k)}{\Delta t} \cdot \mathbf{n}_\Gamma(k-1) + \mathbf{v}_C(k) \cdot \frac{\mathbf{n}_\Gamma(k) - \mathbf{n}_\Gamma(k-1)}{\Delta t} \right]. \quad (26)$$

According to the Newton's law of motion, $\mathbf{v}_{C,d}(k+1)$ is formulated by

$$\mathbf{v}_{C,d}(k+1) = \mathbf{v}_C(k) + \mathbf{a}_C(k) \cdot \Delta t \quad (27)$$

where $\mathbf{v}_{C,d}(k+1)$ denotes the desired velocity of C at the $(k+1)$ th sampling instant, $\mathbf{a}_C(k)$ is the total guidance command of the problem

$$\mathbf{a}_C(k) = u(k)\mathbf{n}_\Gamma(k) + \mathbf{a}_\Gamma(k) \quad (28)$$

where $u(k)$ is the magnitude of control input along $\mathbf{n}_\Gamma(k)$, aimed at diminishing the magnitude of $\mathbf{v}_{C,\perp}(k)$ to zero. Substituting (23) and (27) into (26) yields

$$e_r(k) - e_r(k-1) = E_1 + E_2 u(k) \quad (29)$$

where $E_1 = 2\mathbf{v}_C(k) \cdot \mathbf{n}_\Gamma(k-1)[\mathbf{a}_\Gamma(k) \cdot \mathbf{n}_\Gamma(k) \cdot \Delta t + \mathbf{v}_C(k-1) \cdot (\mathbf{n}_\Gamma(k) - \mathbf{n}_\Gamma(k-1))]$, $E_2 = 2\mathbf{v}_C(k) \cdot \mathbf{n}_\Gamma(k-1) \cdot \mathbf{n}_\Gamma(k) \cdot \Delta t$. The remaining interception time $t_{go,\Gamma}(k)$ on the Γ_k can be approximated as

$$t_{go,\Gamma}(k) = \frac{-r(k-1)}{r_\Gamma(k) - r(k-1)} \Delta t. \quad (30)$$

The combined guidance command is given as

$$\begin{aligned} \mathbf{a}_C(k) = & \left(-\frac{\Delta t e_r(k-1)}{E_2 N t_{go,\Gamma}(k)} - \frac{E_1}{E_2} \right) \mathbf{n}_\Gamma(k) \\ & + \frac{K \cdot \|\mathbf{v}_C(k)\| \Delta q_k}{\Delta t} \hat{\mathbf{a}}_\Gamma(k) \end{aligned} \quad (31)$$

where $0 < N < 1$.

THEOREM 1 With the input of the combined guidance command defined by (31), e_r will converge to zero before the completion of the 2-D guidance process.

PROOF If $n_{jud} = 0$, the geometric relationship is shown in Fig. 4. In Fig. 4(a), one can obtain $r_{VL}(k+1) < r_{VL}(k)$. Furthermore, as deduced from Remark 1, due to the nature of PN, it is evident that with the passage of time, the convergence rate of r_{VL} will progressively accelerate. In scenarios where either Fig. 4(b) or $n_{jud} \neq 0$, as highlighted in Remark 1, the direction of $\mathbf{v}_C(k)$ converges to the VLOS. This signifies that the configuration would transition to the state, as depicted in Fig. 4(a). Then, it is necessary to ensure that $\mathbf{v}_C$ falls on Γ_k before r_Γ converges to zero. By dividing both sides of the (29) by Δt , the derivative of e_r at time t_k is approximated as

$$\dot{e}_r|_{t=t_k} = \frac{E_1}{\Delta t} + \frac{E_2}{\Delta t} u(k). \quad (32)$$

For each sampling time t_k , define an auxiliary variable $e_{r,k} \geq 0$, where $e_{r,k}(k-1) = e_r(k-1)$ and $\dot{e}_{r,k}|_{t \geq t_k} = \dot{e}_r|_{t=t_k}$. Then, for any $t_f(k) > t_k$, we have

$$e_{r,k}(t_f(k)) = e_r(k-1) + \left(\frac{E_1}{\Delta t} + \frac{E_2}{\Delta t} u(k) \right) (t_f(k) - t_k). \quad (33)$$

When $e_{r,k}(t_f(k)) = 0$, $u(k)$ is solved as

$$u(k) = -\frac{\Delta t e_r(k-1)}{E_2 t_f(k) - t_k} - \frac{E_1}{E_2}. \quad (34)$$

It is important to note that when t_k is updated, E_1 , E_2 , and Δt in (32) will also be updated accordingly. By designing $u(k)$ in the form of (34), it is ensured that $e_r(k) - e_r(k-1) \leq 0$ (with equality only when $e_r(k-1) = 0$) for any t_k , and thus e_r will monotonically converge to zero. The remaining convergence time of e_r is approximated as

$$t_{go,\perp}(k) = \frac{-e_r(k-1)}{e_r(k) - e_r(k-1)} \Delta t. \quad (35)$$

To ensure that $t_{go,\perp}(k) < t_{go,\Gamma}(k)$, we can set

$$t_{go,\perp}(k) = N t_{go,\Gamma}(k) \quad (36)$$

where $0 < N < 1$. By combining (29) with (35) and (36), $u(k)$ is solved as

$$u(k) = -\frac{\Delta t e_r(k-1)}{E_2 N t_{go,\Gamma}(k)} - \frac{E_1}{E_2} \quad (37)$$

when $t_{go,\perp}(k) = t_f(k) - t_k$, (37) shares the same form with (34). Therefore, by designing $u(k)$ in the form of (37), it is ensured that e_r will monotonically converge to zero, with a convergence time shorter than the remaining interception time of the 2-D guidance process. Substituting (23) and (37) into (28) yields (31). Then, Theorem 1 is proven. ■

Denote the components of point C^{k+1} 's coordinate on $\mathbf{n}_\Gamma(k)$ as $C_\perp^{k+1}$. Its position vector is denoted by $\mathbf{r}_{C,\perp}(k+1)$, and can be formulated by

$$\mathbf{r}_{C,\perp}(k+1) = \mathbf{r}_C(k) + \Delta t \cdot \mathbf{v}_{C,\perp}(k) + \frac{1}{2} u(k) \Delta t^2 \cdot \mathbf{n}_\Gamma(k). \quad (38)$$

By combining C_Γ^{k+1} and $C_\perp^{k+1}$, the desired coordinate of C^{k+1} is expressed as

$$\mathbf{r}_{C,d}(k+1) = \mathbf{r}_{C,\Gamma}(k+1) + \mathbf{r}_{C,\perp}(k+1) - \mathbf{r}_C(k). \quad (39)$$

REMARK 2 Due to the inherent time delay in the discrete guidance law designed in this section, a certain amount of miss distance (MD) will unavoidably occur at the terminal moment of guidance. However, as long as MD remains within the interception region of the capture net, the successful capture of the target can be ensured.

The normal vector of the plane in which the capture net is located can be represented as

$$\mathbf{n}_C(k) = \frac{\mathbf{r}_2(k) - \mathbf{r}_1(k)}{\|\mathbf{r}_2(k) - \mathbf{r}_1(k)\|} \times \frac{\mathbf{r}_3(k) - \mathbf{r}_2(k)}{\|\mathbf{r}_3(k) - \mathbf{r}_2(k)\|} \quad (40)$$

where the position vectors $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$ will be obtained in Section IV. Then, the distance from the target and the net plane $d_m(k)$ can be calculated as

$$d_m(k) = |\mathbf{n}_C(k) \cdot \mathbf{r}_{VL}(k)|. \quad (41)$$

When $d_m(k) = 0$, the guidance law reaches its terminal state.

Pseudocode for the proposed guidance law is given in Algorithm 1.

REMARK 3 From (28), the guidance command $\mathbf{a}_C(k)$ contains two components, $\mathbf{a}_\Gamma(k)$ and $u(k)\mathbf{n}_\Gamma(k)$. The component $\mathbf{a}_\Gamma(k)$ lies in Γ_k , and perpendicular to $\mathbf{v}_{C,\Gamma}(k)$, which

Algorithm1: PN-like Discrete-Time 3-D Guidance Law.

Input: $\mathbf{r}_C(k)$, $\mathbf{r}_T(k)$, $\mathbf{v}_i(k)$, $i = 1, 2, \dots, n$
Output: $\mathbf{r}_{C,d}(k+1)$, $\mathbf{v}_{C,d}(k+1)$, $\mathbf{a}_C(k)$

```

1  $k = 1$ ;
2 while  $d_m(k) > 0$  do
3   Solve  $n_{jud}$ ;
4   if  $n_{jud} = 0$  then
5     Solve  $\mathbf{r}_{C,r}(k+1)$  by (11)-(18);
6   else
7     Solve  $\mathbf{r}_{C,r}(k+1)$  by (18)-(21);
8   Get  $\mathbf{a}_C(k)$  by (31);
9   Solve  $\mathbf{r}_{C,d}(k+1)$  and  $\mathbf{v}_{C,d}(k+1)$  by (27)
    and (39);
10   $k = k + 1$ ;

```

is similar to PPN, as shown in (23), while $\mathbf{u}(k)\mathbf{n}_{\Gamma_k}(k)$ is perpendicular to Γ_k , as shown in (31). Therefore, the total guidance command $\mathbf{a}_C(k)$ may not be perpendicular to $\mathbf{v}_C$, until $\mathbf{v}_C(k) = \mathbf{v}_{C,r}(k)$, $u(k) = 0$, which means that the guidance command degenerates into a 2-D PPN guidance law. Therefore, we refer to $\mathbf{a}_C(k)$ as a 3-D PN-like guidance law.

IV. ORIENTATING STRATEGY FOR NET PLANE

A. Solution of UAV Formation

As depicted in Fig. 2, all the UAVs are arranged counterclockwise. Among the UAV formation, the two UAVs with the lowest altitude are defined as P_1 and P_2 . In order to determine the desired positions of all the net-carrying UAVs, it is imperative to solve the desired positions of P_1 and P_2 at once as reference utilizing $\mathbf{r}_{C,d}(k+1)$, which has been solved in Section III. Denote the desired position of the i th UAV at the $(k+1)$ th sampling instant as P_{id}^{k+1} . Then, one can obtain

$$\begin{cases} z_{1,d}(k+1) = z_{2,d}(k+1) \\ z_C(k+1) > z_{1,d}(k+1). \end{cases} \quad (42)$$

To adjust the orientation of the net, the VLOS vector should be perpendicular to the plane formed by P_{1d}^{k+1} , P_{2d}^{k+1} , and C^{k+1} [10]. Then, one can obtain

$$\begin{aligned} & \frac{(\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{1,d}(k+1)) \times (\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{2,d}(k+1))}{\|(\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{1,d}(k+1)) \times (\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{2,d}(k+1))\|} \\ &= \begin{bmatrix} \cos \theta_{VL}(k+1) \sin \phi_{VL}(k+1) \\ \cos \theta_{VL}(k+1) \cos \phi_{VL}(k+1) \\ \sin \theta_{VL}(k+1) \end{bmatrix}. \end{aligned} \quad (43)$$

The distance between P_{1d}^{k+1} and P_{2d}^{k+1} has been defined as L , and thus, one can obtain

$$\|\mathbf{r}_{1,d}(k+1) - \mathbf{r}_{2,d}(k+1)\| = L. \quad (44)$$

Next, consider the triangle constructed by P_{1d}^{k+1} , P_{2d}^{k+1} , and C^{k+1} . According to the geometric relationship, one can get

$$\begin{cases} l = \sin\left(\frac{\pi}{2} - \frac{\pi}{n}\right) \cdot \frac{L}{\sin\left(\frac{2\pi}{n}\right)} & (n \geq 3, n \in \mathbb{Z}) \\ \|\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{1,d}(k+1)\| = l \\ \|\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{2,d}(k+1)\| = l \end{cases} \quad (45)$$

where l is the distance between the vertex of a regular polygon composed of net-carrying UAVs and its geometric center. Then, the desired positions of P_{1d}^{k+1} and P_{2d}^{k+1} can be expressed by

$$\begin{cases} \mathbf{r}_{1,d}(k+1) = G_1(\mathbf{r}_{C,d}(k+1), L, n) \\ \mathbf{r}_{2,d}(k+1) = G_2(\mathbf{r}_{C,d}(k+1), L, n) \end{cases} \quad (46)$$

where $G_1(\mathbf{r}_{C,d}(k+1), L, n)$ and $G_2(\mathbf{r}_{C,d}(k+1), L, n)$ are lengthy, but can be readily derived by solving (42)–(45). The solution of the desired position of the i th UAV is similar to that of P_{1d}^{k+1} and P_{2d}^{k+1} . The constraint equations are written as

$$\begin{cases} \frac{(\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{1,d}(k+1)) \times (\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{2,d}(k+1))}{\|(\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{1,d}(k+1)) \times (\mathbf{r}_{C,d}(k+1) - \mathbf{r}_{2,d}(k+1))\|} \\ = \begin{bmatrix} \cos \theta_{VL}(k+1) \sin \phi_{VL}(k+1) \\ \cos \theta_{VL}(k+1) \cos \phi_{VL}(k+1) \\ \sin \theta_{VL}(k+1) \end{bmatrix} \\ \|\mathbf{r}_{(i+1),d}(k+1) - \mathbf{r}_{C,d}(k+1)\| = l \\ \|\mathbf{r}_{(i+1),d}(k+1) - \mathbf{r}_{i,d}(k+1)\| = L. \end{cases} \quad (47)$$

In addition, considering the triangle constructed by $P_{(i-1)d}^{k+1}$, P_{id}^{k+1} and $P_{(i+1)d}^{k+1}$, one can obtain

$$\begin{cases} h = \sin\left(\pi - \frac{2\pi}{n}\right) \cdot \frac{L}{\sin\left(\frac{\pi}{n}\right)} \\ \|\mathbf{r}_{(i+1),d}(k+1) - \mathbf{r}_{(i-1),d}(k+1)\| = h. \end{cases} \quad (48)$$

Then, the desired position coordinates of the $(i+1)$ th UAV can be expressed as

$$\mathbf{r}_{(i+1),d}(k+1) = G_{i+1}(\mathbf{r}_{C,d}(k+1), L, n) \quad (49)$$

where $G_{i+1}(\mathbf{r}_{C,d}(k+1), L, n)$ is lengthy, but can be readily obtained by solving (47) and (48).

Notice that the strategy is under discrete-time process, $\mathbf{v}_{i,d}(k+1)$ and $\mathbf{a}_{i,d}(k+1)$ can be expressed by backward difference and central difference of position, i.e.,

$$\begin{cases} \mathbf{v}_{i,d}(k+1) = \frac{\mathbf{r}_{i,d}(k+1) - \mathbf{r}_{i,d}(k)}{\Delta t} \\ \mathbf{a}_{i,d}(k+1) = \frac{\mathbf{r}_{i,d}(k+1) - 2\mathbf{r}_{i,d}(k) + \mathbf{r}_{i,d}(k-1)}{2\Delta t^2}. \end{cases} \quad (50)$$

One can represent the net angle as θ_{Γ_k} , which describes the angle between the direction of motion of C and the plane formed by the n net-carrying UAVs

$$\theta_{\Gamma_k}(k) = \frac{\pi}{2} - \frac{\mathbf{n}_p(k) \cdot \mathbf{n}_C(k)}{\|\mathbf{n}_p(k)\| \cdot \|\mathbf{n}_C(k)\|}$$

where $\mathbf{n}_C(k) = \mathbf{r}_C(k) - \mathbf{r}_C(k-1)$ and $\mathbf{n}_p(k) = (\mathbf{r}_1(k) - \mathbf{r}_2(k)) \times (\mathbf{r}_3(k) - \mathbf{r}_2(k))$. The interception strategy performs optimally when $\theta_{\Gamma_k}(k) = \pi/2$.

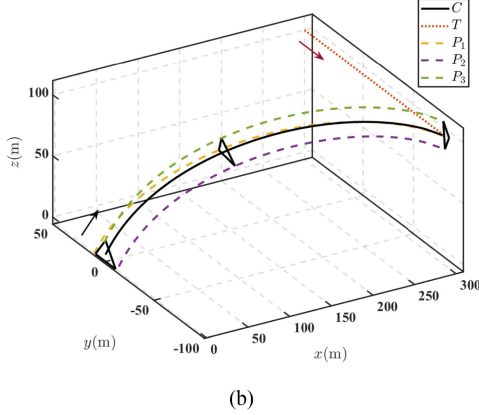
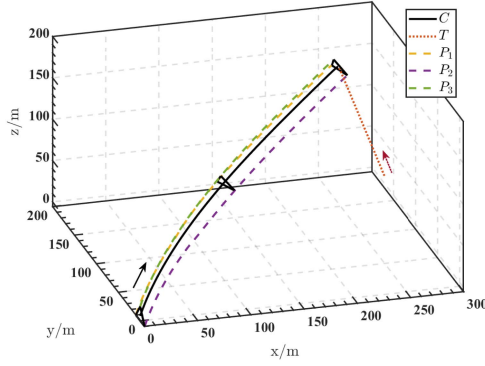


Fig. 6. Trajectories for the scenario (i). (a) $n_{jud} = 0$. (b) $n_{jud} \neq 0$.

B. Control of Net-Carrying UAV

Transforming (2) into discrete form yields

$$\begin{cases} \mathbf{r}_i(k+1) = \mathbf{r}_i(k) + \mathbf{v}_i(k) \cdot \Delta t \\ \mathbf{v}_i(k+1) = \mathbf{v}_i(k) + \mathbf{U}_i(k) \cdot \Delta t + \mathbf{d}_i(k). \end{cases} \quad (51)$$

Define $\tilde{\mathbf{r}}_i(k) = \mathbf{r}_i(k) - \mathbf{r}_{i,d}(k)$, $\tilde{\mathbf{v}}_i(k) = \mathbf{v}_i(k) - \mathbf{v}_{i,d}(k)$ as the position and velocity tracking error, respectively. Rearrange (51) to obtain a nominal system as

$$\begin{cases} \tilde{\mathbf{r}}_i(k+1) = \mathbf{r}_i(k) + \mathbf{v}_i(k) \cdot \Delta t - \mathbf{r}_{i,d}(k+1) \\ \tilde{\mathbf{v}}_i(k+1) = \mathbf{v}_i(k) + \mathbf{U}_i(k) \cdot \Delta t + \mathbf{d}_i(k) - \mathbf{v}_{i,d}(k+1). \end{cases} \quad (52)$$

Inspired by [34] and [38], to compensate the disturbance $\mathbf{d}_i(k) = \mathbf{D}_i(k)\Delta t$, an estimated term $\hat{\mathbf{u}}_{i,d}$ can be designed as the output of the following dynamics estimator:

$$\hat{\mathbf{u}}_{i,d}(k) = \mathbf{u}_{i,v}(k) - \mathbf{u}_{i,u}(k) \quad (53)$$

where $\mathbf{u}_{i,v}$ and $\mathbf{u}_{i,u}$ are the filtered signals by filtering $\tilde{\mathbf{v}}_i$ and $\mathbf{v}_i + \mathbf{U}_i \cdot \Delta t - \mathbf{v}_{i,d}$ through the following first-order filters:

$$\mathbf{u}_{i,v}(k+1) = \alpha_{i,r} \tilde{\mathbf{v}}(k+1) + (1 - \alpha_{i,r}) \mathbf{u}_{i,v}(k) \quad (54)$$

$$\begin{aligned} \mathbf{u}_{i,u}(k+1) &= \alpha_{i,r} (\mathbf{v}_i(k) + \mathbf{U}_i(k) \cdot \Delta t - \mathbf{v}_{i,d}(k+1)) \\ &\quad + (1 - \alpha_{i,r}) \mathbf{u}_{i,u}(k) \end{aligned} \quad (55)$$

where $0 < \alpha_{i,r} < 1$ is the filter parameter. The initial states $\mathbf{u}_{i,v}(k) = [0, 0, 0]^T$ and $\mathbf{u}_{i,u}(k) = [0, 0, 0]^T$.

THEOREM 2 For system (52) with estimator (53), the estimated error $\tilde{\mathbf{u}}_{i,d}(k) = \hat{\mathbf{u}}_{i,d}(k) - \mathbf{d}_i(k)$ is uniformly ultimately bounded.

PROOF Regarding the design procedure of the estimator, $\tilde{\mathbf{u}}_{i,d}(k)$ can be interpreted as the filtered version of the disturbance $\mathbf{d}_i(k)$ given by $\hat{\mathbf{u}}_{i,d}(k+1) = \alpha_{i,v} \mathbf{d}_i(k) + (1 - \alpha_{i,v}) \hat{\mathbf{u}}_{i,d}(k)$. Then, we have

$$\begin{aligned} \tilde{\mathbf{u}}_{i,d}(k+1) &= \hat{\mathbf{u}}_{i,d}(k+1) - \mathbf{d}_i(k) \\ &= (1 - \alpha_{i,r}) \tilde{\mathbf{u}}_{i,d}(k) - \Delta \mathbf{d}_i(k) \end{aligned} \quad (56)$$

where $\Delta \mathbf{d}_i(k) = \mathbf{d}_i(k+1) - \mathbf{d}_i(k)$. Consider the Lyapunov function $V_d(k) = \tilde{\mathbf{u}}_{i,d}(k)^T \tilde{\mathbf{u}}_{i,d}(k)$, the difference of the Lyapunov function is given by

$$\begin{aligned} \Delta V_d(k+1) &= V_d(k+1) - V_d(k) \\ &= [(1 - \alpha_{i,r}) \tilde{\mathbf{u}}_{i,d}(k) + \Delta \mathbf{d}_i(k)]^2 - \tilde{\mathbf{u}}_{i,d}(k)^2 \\ &= (\alpha_{i,r}^2 - 2\alpha_{i,r}) \tilde{\mathbf{u}}_{i,d}(k)^2 + 2(1 - \alpha_{i,r}) \tilde{\mathbf{u}}_{i,d}(k) \Delta \mathbf{d}_i(k) \\ &\leq (\alpha_{i,r}^2 - \alpha_{i,r}) \tilde{\mathbf{u}}_{i,d}(k)^2 + \left(1 + \frac{(1 - \alpha_{i,r})^2}{\alpha_{i,r}}\right) \sigma_d^2 \Delta t^2 \end{aligned} \quad (57)$$

where the conclusion $\tilde{\mathbf{u}}_{i,d}(k) \Delta \mathbf{d}_i(k) \leq \frac{\alpha_{i,r}}{2(1 - \alpha_{i,r})} \tilde{\mathbf{u}}_{i,d}(k)^2 + \frac{(1 - \alpha_{i,r})}{2\alpha_{i,r}} \Delta \mathbf{d}_i(k)^2$ and Assumption 2 are used.

Therefore, as in a similar analysis procedure found in [39], it suggests that $\Delta V_d(k+1) < 0$ under the following conditions:

$$\|\tilde{\mathbf{u}}_{i,d}(k)\| \leq \frac{\sigma_d \Delta t}{\alpha_{i,r}} \sqrt{\frac{\alpha_{i,r}^2 - 2\alpha_{i,r} + 2}{1 - \alpha_{i,r}}}$$

Hence, it can be concluded that $\tilde{\mathbf{u}}_{i,d}(k)$ is uniformly ultimately bounded if $0 < \alpha_{i,r} < 1$. In particular, $\|\tilde{\mathbf{u}}_{i,d}(k)\| \rightarrow 0$ holds for $\sigma_d \rightarrow 0$. ■

The pseudocontrol variables $\mathbf{U}_i$ can be designed based on discrete-time PD control law with feedforward as

$$\mathbf{U}_i(k) = -k_p \tilde{\mathbf{r}}(k) - k_d \tilde{\mathbf{v}}(k) + \mathbf{a}_{i,d}(k) - \frac{\hat{\mathbf{u}}_{i,d}(k)}{\Delta t}. \quad (58)$$

Subsequently, substituting (58) into (3), the actual control command $\phi_{b_i}(k)$, $n_{g_i}(k)$, and $T_{h_i}(k)$ can be obtained.

V. NUMERICAL SIMULATIONS

Numerical simulations are provided to demonstrate the effectiveness of the interception strategy developed in Sections III and IV.

In this article, the drag in (1) is expressed by [40]

$$D_{g_i} = 0.5 \rho (V_i - V_{w_i})^2 SC_{D0} + \frac{2k_g k_n^2 n_{g_i}^2 m_i^2}{\rho (V_i - V_{w_i})^2 S}$$

where the constants used are wing area $S = 1.37 \text{ m}^2$, zero-lift drag coefficient $C_{D0} = 0.02$, load-factor effectiveness $k_n = 1$, induced drag coefficient $k_g = 0.1$, the atmospheric density $\rho = 1.225 \text{ kg/m}^3$, the mass of the UAV $m_i = 20 \text{ kg}$,

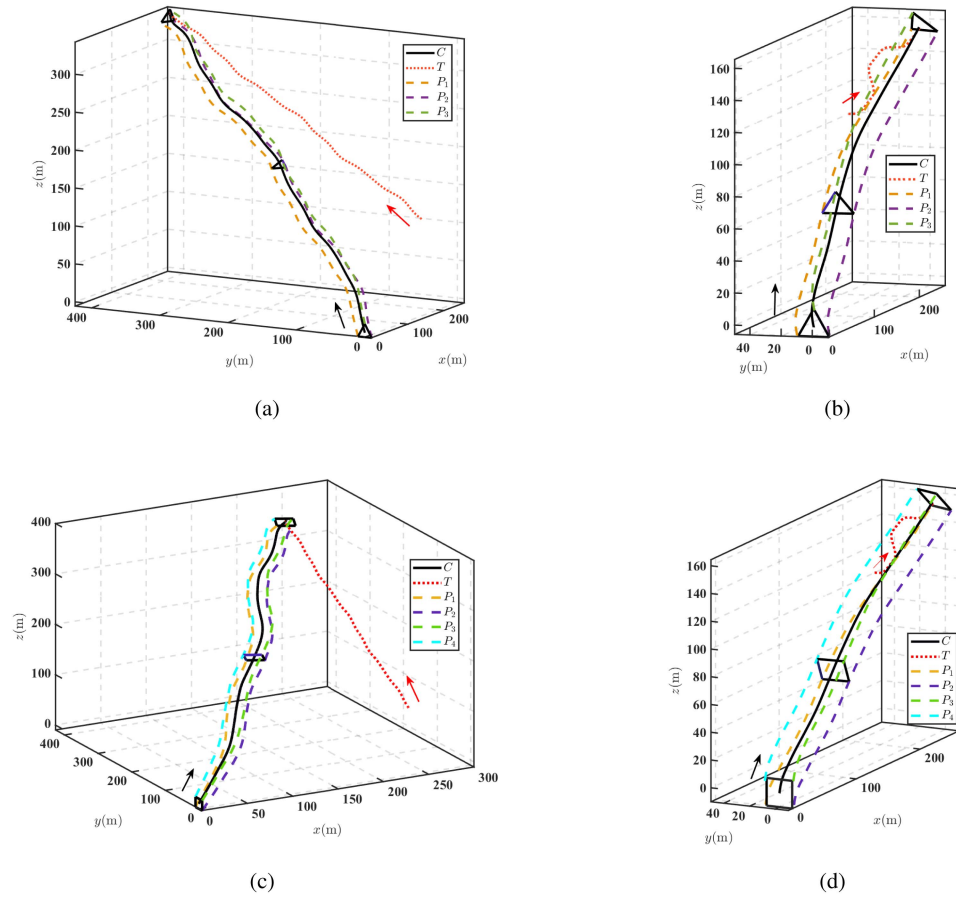


Fig. 7. Trajectories for the scenario (ii). (a) $n = 3$, $n_{jud} = 0$. (b) $n = 3$, $n_{jud} \neq 0$. (c) $n = 4$, $n_{jud} = 0$. (d) $n = 4$, $n_{jud} \neq 0$.

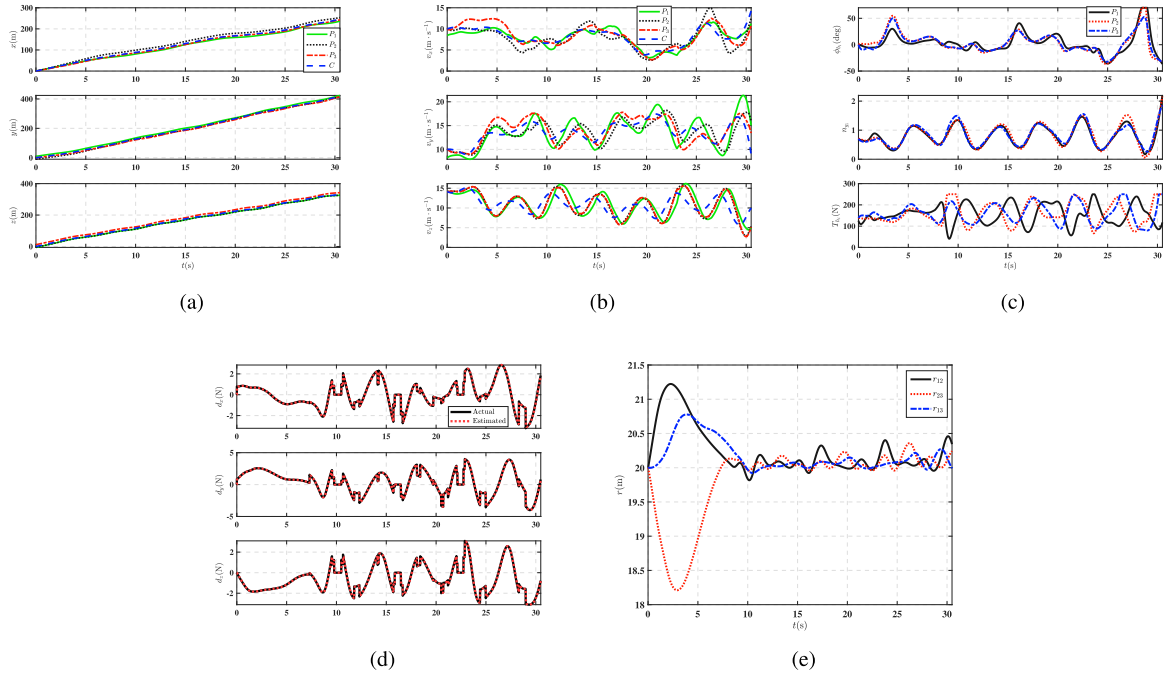


Fig. 8. Detailed plots for the scenario (ii) under $n = 3$. (a) Position. (b) Velocity. (c) Control variables. (d) Estimation of the external disturbance. (e) Relevant inter-UAV distances.

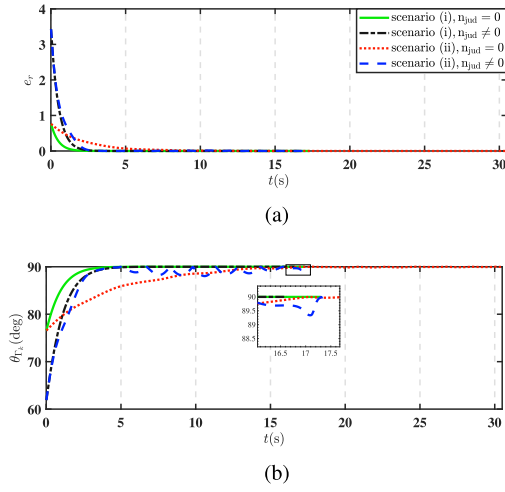


Fig. 9. Major process variables for scenario (i) and (ii). (a) e_r . (b) θ_{Γ_k} .

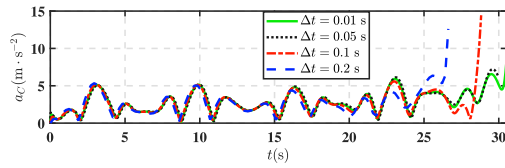


Fig. 10. Magnitude of the guidance command.

and gravitational coefficient $g = 9.81 \text{ m/s}^2$. The gust model $V_{\omega_i} = V_{\omega_i, \text{nom}} + V_{\omega_i, \text{tan}}$ is scaled based on [41] and varies according to the altitude z_i . The simulated normal wind shear is given by

$$V_{\omega_i, \text{nom}} = 0.215V_m \log(z_i) + 0.285V_m$$

where $V_m = 5.0 \text{ m/s}$ is the mean wind speed at an altitude of 300 m. The turbulence part of the wind gust $V_{\omega_i, \text{tan}}$ has a Gaussian distribution with a zero mean and a standard derivation of $0.09V_m$. Other parameters are set as $k_c = 2.5 \text{ N/m}$, $c = 0.8 \text{ N}\cdot\text{m/s}^2$, $k_p = 2.2$, $k_d = 1$, and $\alpha_{ir} = 0.5$. The constraints on the control variables are $T_{hi} < 250 \text{ N}$, $-0.5 \leq n_{gi} \leq 2.5$, and $-70^\circ \leq \phi_{bi} \leq 70^\circ$.

Assume $n = 3$, $L = 20 \text{ m}$, $\Delta t = 0.01 \text{ s}$ in the simulation, that is, P_1 , P_2 , and P_3 arrange in a form of an equilateral triangle with a side length of 20 m, where P_1P_2 is parallel to the horizontal plane. The initial coordinates of P_1 , P_2 , and P_3 are $(0, 10, -10\sqrt{3}/3) \text{ m}$, $(0, -10, -10\sqrt{3}/3) \text{ m}$, and $(0, 0, 20\sqrt{3}/3) \text{ m}$, respectively. In addition, to demonstrate the generality of the proposed algorithm, another condition is considered: $n = 4$, and the initial coordinates of P_1 , P_2 , P_3 , and P_4 are $(0, 10, -10) \text{ m}$, $(0, -10, -10) \text{ m}$, $(0, -10, 10) \text{ m}$, and $(0, 10, 10) \text{ m}$, respectively. The intruder's initial coordinate is $(250, 50, 100) \text{ m}$. The initial velocity of the net-carrying UAVs is 20 m/s , whereas the intruder's initial velocity is $[4\sqrt{3}, 4, 0]^T \text{ m/s}$ under the condition of $n_{jud} = 0$, where the same variable is $[0, -8, 0]^T \text{ m/s}$ under the condition of $n_{jud} \neq 0$. The guidance gain is set as $K = 3$.

Two scenarios are considered here. In scenario (i), the intruder T moves with a constant velocity. In scenario (ii), T moves with a time-varying acceleration. Figs. 6 and 7 show the trajectories of UAVs and intruder in the above two

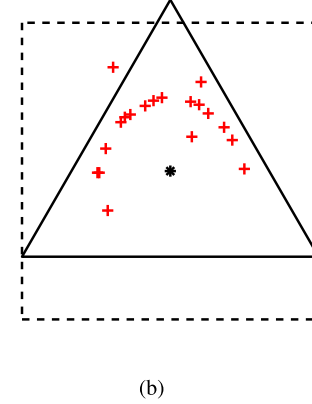
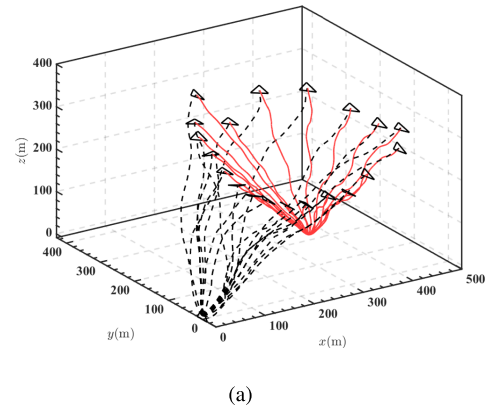


Fig. 11. (a) Trajectories of the target and the capture net with random initial heading angles. (b) Interception point on the capture net.

scenarios, where the black solid lines trace the trajectories of the net boundaries and the net center C , and the dashed lines represent the flight paths of the individual UAVs. The intruder's trajectories are represented by red dotted lines. The black triangles indicate the net plane's position during the initial, intermediate, and terminal sampling instants. The UAVs perform the change of net angle during the intercept process. In Fig. 6, the intruder maintains a constant velocity, and the capture net is capable of intercepting the intruder with the suitable orientation in scenarios where $n_{jud} \neq 0$ and $n_{jud} = 0$. In Fig. 7, the intruder moves with an acceleration of $a_T = [5 \sin(2t), 5 \sin(2t) \cos t, 5 \sin(2t)]^T \text{ m/s}^2$. It is evident that the proposed interception strategy is capable for capturing maneuvering intruder with an appropriate net angle. Fig. 7(c) and (d) show the trajectories which are under the condition of $n = 4$.

Detailed plots for the scenario (ii), where T moves with a time-varying acceleration under the condition of $n_{jud} = 0$ are shown in Fig. 8. The time history of position coordinates of P_1 , P_2 , P_3 and C are shown in Fig. 8(a). One can notice that the z coordinate of P_1 and P_2 are always the same because P_1P_2 is always parallel to the horizontal plane. The velocity components of P_1 , P_2 , P_3 and C are shown in Fig. 8(b). Fig. 8(c) and (d) show the control variables and the estimated disturbance of the net-carrying UAVs. It is evident that the proposed dynamic estimator is capable of estimating external disturbance, and the UAV

TABLE I
Miss Distance for Scenario (ii) Corresponding to
Different Sampling Steps

t (s)	0.01	0.05	0.1	0.2
MD (m)	0.089	0.106	10.875	14.314

can maneuver to intercept the target while satisfying the control constraints. Fig. 8(e) illustrates the time histories of the relevant inter-UAV distances. With an expected distance between UAVs of $L = 20$ m, it can be observed that, after overshooting, the error of relevant inter-UAV distances do not exceed 0.5 m. The maximum stretch length of the capture net's edge does not exceed 10% of its original length.

The plots of major process variables for the two scenarios under the condition of $n = 3$ are provided in Fig. 9. The time history of e_r is shown in Fig. 9(a), from which we can infer that P_1 , P_2 , and P_3 are moving toward the instantaneous guidance plane Γ_k gradually until the direction of UAVs' velocity falls on it. In Fig. 9(d), the time history of the net angle θ_{Γ_k} is revealed, and it is evident that the net angles are gradually close to 90° before $k_n = 1$. From these sets of plot, one can observe that though the maneuvering characteristics of the intruder may take effect on the guidance process, the proposed interception strategy can still accomplish the guidance objectives.

The simulation results illustrating the impact of different sampling steps ($\Delta t = 0.01, 0.05, 0.1, 0.2$ s) on the guidance performance [corresponding to scenario (ii), $n = 3$, $n_{\text{jud}} = 0$] are presented in Table I and Fig. 10. Table I shows the MD corresponding to different sampling steps. Fig. 10 shows the time history of the magnitude of the guidance command a_C . It can be observed that as the sampling steps decreases, the required overload at the end of the guidance process increases, leading to larger MD. This is because as r_{VL} decreases, the change in LOS angle becomes more violent, thus requiring larger acceleration commands.

An additional set of simulation results is presented for a maneuvering target [corresponding to scenario (ii)] with intruder and capture net having randomly chosen initial heading angles. These results are shown in Fig. 11. Fig. 11(a) illustrates the distinct spatial trajectories of both the target and the net center. In Fig. 11(b), the interception point is marked by a red "+" on the capture net, the center of the net is marked by a black "*", where solid lines represent $n = 3$ and dotted lines represent $n = 4$. It can be observed that even when there is a certain MD between the target and the center of the net, the capture net can still successfully intercept the target. Moreover, under the condition of constant inter-UAV spacing, an increase in the number of net-carrying UAVs leads to a larger interceptable area.

VI. CONCLUSION

The capture problem against an aerial intruder by n net-carrying UAVs has been investigated in this article. A 3-D discrete-time PN-like guidance law is presented to minimize the distance between the capture net and the

intruder. Importantly, it achieves this objective solely by utilizing the position error vectors measured between each net-carrying UAV and the intruder. Furthermore, a net-plane orientating strategy considering the dynamics of the UAVs and net tension disturbance is proposed to guarantee that the net can capture the intruder properly. Simulation results demonstrate that the net-carrying UAVs can effectively capture the intruder with an the appropriate net orientation. There are several potential directions for the future work in this problem. The first would be development of the capture strategy on the recovery process, which aims to change the formation of the UAVs to wrap the intruder by the net. Moreover, the testing of the algorithm on an experimental platform is another avenue of research which needs to be pursued.

REFERENCES

- [1] K. Klausen, T. I. Fossen, and T. A. Johansen, "Autonomous recovery of a fixed-wing UAV using a net suspended by two multirotor UAVs," *J. Field Robot.*, vol. 35, no. 5, pp. 717–731, 2018.
- [2] M. R. Aagaah, E. M. Ficanha, and N. Mahmoudian, "Drone having drone-catching feature," US Patent 10,005,556, Jun. 26 2018.
- [3] H. J. Kim et al., "Fully autonomous vision-based net-recovery landing system for a fixed-wing UAV," *IEEE/ASME Trans. Mechatron.*, vol. 18, no. 4, pp. 1320–1333, Aug. 2013.
- [4] M. Diehl and M. Hornung, "Conceptual design of an interceptor UAV," in *Proc. 17th AIAA Aviation Technol., Integration, Operations Conf.*, 2017, Art. no. 3587.
- [5] K. Klausen, J. B. Moe, J. C. van den Hoorn, A. Gomola, T. I. Fossen, and T. A. Johansen, "Coordinated control concept for recovery of a fixed-wing UAV on a ship using a net carried by multirotor UAVs," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2016, pp. 964–973.
- [6] J. Rothe, M. Strohmeier, and S. Montenegro, "A concept for catching drones with a net carried by cooperative UAVs," in *Proc. IEEE Int. Symp. Safety, Secur., Rescue Robot.*, 2019, pp. 126–132.
- [7] S. Tolman and R. W. Beard, "Counter UAS using a formation controlled dragnet," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2017, pp. 1665–1672.
- [8] J. Kim, "Three-dimensional discrete-time controller to intercept a targeted UAV using a capture net towed by multiple aerial robots," *IET Radar, Sonar Navigation*, vol. 13, no. 5, pp. 682–688, 2019.
- [9] D. Lee, S. Kim, and J. Suk, "Formation flight of unmanned aerial vehicles using track guidance," *Aerosp. Sci. Technol.*, vol. 76, pp. 412–420, 2018.
- [10] A. Chakravarthy and D. Ghose, "Collision cone-based net capture of a swarm of unmanned aerial vehicles," *J. Guid., Control, Dyn.*, vol. 43, no. 9, pp. 1688–1710, 2020.
- [11] A. Chakravarthy and D. Ghose, "Cooperative pursuit guidance to surround intruder swarms using collision cones," *J. Aerosp. Inf. Syst.*, vol. 17, no. 8, pp. 454–469, 2020.
- [12] A. Chakravarthy and D. Ghose, "Intercepting intruder UAV swarms with a net manipulated by a team of UAVs," in *Proc. AIAA Scitech Forum*, 2021, Art. no. 1767.
- [13] J. Brown and N. Raj, "Guidance law for a surveillance UAV swarm tracking a high capability malicious UAV," in *Proc. IEEE Asia Pacific Conf. Wireless Mobile*, 2021, pp. 226–232.
- [14] C.-Y. Li and W.-X. Jing, "Geometric approach to capture analysis of PN guidance law," *Aerosp. Sci. Technol.*, vol. 12, no. 2, pp. 177–183, 2008.
- [15] F. Tyan, "Analysis of 3D PPN guidance laws for nonmaneuvering target," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 4, pp. 2932–2943, Oct. 2015.
- [16] K.-B. Li, H.-S. Shin, A. Tsourdos, and M.-J. Tahk, "Capturability of 3D PPN against lower-speed maneuvering target for homing phase," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 1, pp. 711–722, Feb. 2020.

- [17] D. Zhou and B. Xu, "Adaptive dynamic surface guidance law with input saturation constraint and autopilot dynamics," *J. Guid., Control, Dyn.*, vol. 39, no. 5, pp. 1155–1162, 2016.
- [18] G. Weiss and I. Rusnak, "All-aspect three-dimensional guidance law based on feedback linearization," *J. Guid., Control, Dyn.*, vol. 38, no. 12, pp. 2421–2428, 2015.
- [19] D. Liu, H. Liu, and K. P. Valavanis, "Fully distributed time-varying formation control for multiple uncertain missiles," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 3, pp. 1646–1656, Jun. 2021.
- [20] W. Dong, C. Wang, J. Wang, Z. Zuo, and J. Shan, "Fixed-time terminal angle-constrained cooperative guidance law against maneuvering target," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 2, pp. 1352–1366, Apr. 2022.
- [21] W. Dong, C. Wang, J. Liu, J. Wang, and M. Xin, "Three-dimensional vector guidance law with impact time and angle constraints," *J. Franklin Inst.*, vol. 360, no. 2, pp. 693–718, 2023.
- [22] W. Dong, C. Wang, J. Wang, and M. Xin, "Varying-gain proportional navigation guidance for precise impact time control," *J. Guid., Control, Dyn.*, vol. 46, no. 3, pp. 535–552, 2023.
- [23] W. Dong, C. Wang, J. Wang, H. Son, and M. Xin, "Unified method for field-of-view-limited homing guidance," *J. Guid., Control, Dyn.*, vol. 45, no. 8, pp. 1415–1434, 2022.
- [24] S. Jana and D. Ghose, "A three-dimensional guidance law for a dual interceptor system," in *Proc. AIAA SCITECH Forum*, 2023, Art. no. 2163.
- [25] S. He, W. Wang, and J. Wang, "Discrete-time super-twisting guidance law with actuator faults consideration," *Asian J. Control*, vol. 19, no. 5, pp. 1854–1861, 2017.
- [26] I. Rusnak, "Discrete optimal guidance for high-order missile and maneuvering target," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 27, no. 6, pp. 870–872, Nov. 1991.
- [27] D. Zhou, S. Sun, J. Y. Zhou, and K. L. Teo, "A discrete sliding-mode guidance law," *J. Dyn. Syst., Meas., Control*, vol. 137, no. 2, 2014, Art. no. 024501.
- [28] R. Gitizadeh, I. Yaesh, and J. Ben-Asher, "Discrete-time optimal guidance," *J. Guid., Control, Dyn.*, vol. 22, no. 1, pp. 171–175, 1999.
- [29] N. Lechevin and C. A. Rabbath, "Robust discrete-time proportional-derivative navigation guidance," *J. Guid., Control, Dyn.*, vol. 35, no. 3, pp. 1007–1013, 2012.
- [30] S. Gao, "Simulation on missile ideal trajectory under proportional navigation," *Comput. Eng. Des.*, vol. 24, no. 8, pp. 66–68, 2003.
- [31] B. Kada and Y. Ghazzawi, "Robust PID controller design for an UAV flight control system," in *Proc. World Congr. Eng. Comput. Sci.*, vol. 2, 2011, pp. 1–6.
- [32] H. Lee, S. Yoon, and S. Han, "Robust attitude control scheme for constantly spinning small-sized quadrotor in the presence of imbalance characteristics," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 2, pp. 1405–1415, Apr. 2022.
- [33] K. Xia, S.-M. Lee, W. Chung, Y. Zou, and H. Son, "Moving target landing of a quadrotor using robust optimal guaranteed cost control," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 3, pp. 819–821, Mar. 2023.
- [34] K. Xia, M. Shin, W. Chung, M. Kim, S. Lee, and H. Son, "Landing a quadrotor UAV on a moving platform with sway motion using robust control," *Control Eng. Pract.*, vol. 128, 2022, Art. no. 105288.
- [35] P. K. Menon, G. D. Sweriduk, and B. Sridhar, "Optimal strategies for free-flight air traffic conflict resolution," *J. Guid., Control, Dyn.*, vol. 22, no. 2, pp. 202–211, 1999.
- [36] Q. Chen, Q. Zhang, Q. Gao, Z. Feng, Q. Tang, and G. Zhang, "Design and optimization of a space net capture system based on a multi-objective evolutionary algorithm," *Acta Astronautica*, vol. 167, pp. 286–295, 2020.
- [37] S. Qu, X. Xia, and J. Zhang, "Dynamics of discrete-time sliding-mode-control uncertain systems with a disturbance compensator," *IEEE Trans. Ind. Electron.*, vol. 61, no. 7, pp. 3502–3510, Jul. 2014.
- [38] J. Na et al., "Air-fuel ratio control of spark ignition engines with unknown system dynamics estimator: Theory and experiments," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 2, pp. 786–793, Mar. 2021.
- [39] J. Sarangapani, *Neural Network Control of Nonlinear Discrete-Time Systems*. Boca Raton, FL, USA: CRC Press, 2018.
- [40] Y. Xu, "Nonlinear robust stochastic control for unmanned aerial vehicles," *J. Guid., Control, Dyn.*, vol. 32, no. 4, pp. 1308–1319, 2009.
- [41] D. File, "Advisory circular," Jun. 2023.



Jianan Wang (Senior Member, IEEE) received the B.S. degree from the Beijing Jiaotong University, Beijing, China, in 2004, the M.S. degree from Beijing Institute of Technology, Beijing, China, in 2007, both in control science and engineering, and the Ph.D. degree in aerospace engineering with Mississippi State University, Starkville, MS, USA, in 2011.

After graduation, he held two Postdoctoral positions with the University of Central Florida, Orlando, FL, USA, and University of Manchester, Manchester, U.K. He is currently a Full Professor with the School of Aerospace Engineering, Beijing Institute of Technology. He has authored or coauthored two textbooks and two research monographs and over 100 papers which were published in most top control journals, such as IEEE TRANSACTIONS ON AUTOMATIC CONTROL, *Automatica*, *AIAA Journal of Guidance, Control, and Dynamics*, and IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS, etc. His research interests include cooperative detection, guidance and control of multiple flight vehicles.

Dr. Wang is a Senior Member of American Institute of Aeronautics and Astronautics. He also served as an Associate Editor for *Unmanned Systems*.



Longze Zhao received the B.S. degree in mechanical engineering from the Northeastern University, Shenyang, China, in 2019. He is currently working toward the Ph.D. degree in aeronautical and astronautical science and technology with the Beijing Institute of Technology, Beijing, China.

His research interests include unmanned aerial vehicle control and navigation system.



Kewei Xia received the Ph.D. degree in control theory and applications from Beihang University, Beijing, China, in 2017.

He was a Postdoctoral Fellow with the Yonsei University Observatory, Yonsei University, Seoul, South Korea, from 2017 to 2018, and a Postdoctoral Researcher with the Department of Mechanical Engineering, Ulsan National Institute of Science and Technology, Ulsan, South Korea, from 2019 to 2021. He is currently an Assistant Professor with the Advanced Research

Institute of Multidisciplinary Sciences, Beijing Institute of Technology, Beijing, China. His research interests include unmanned aerial vehicle control and multiagent control.

Dr. Xia currently serves as an Associate Editor for *Journal of Vibration and Control*.



Chunyan Wang received the M.Sc. degree in electrical and electronic engineering from the University of Greenwich, London, U.K., in 2012, and the Ph.D. degree in control systems from the University of Manchester, Manchester, U.K., in 2016.

He was a Research Associate with the School of Electrical and Electronic Engineering, University of Manchester, from 2016 to 2018. He is currently an Associate Professor with the School of Aerospace Engineering, Beijing Institute of Technology, Beijing, China. His research interests include cooperative guidance and control, UAVs, and robust control.